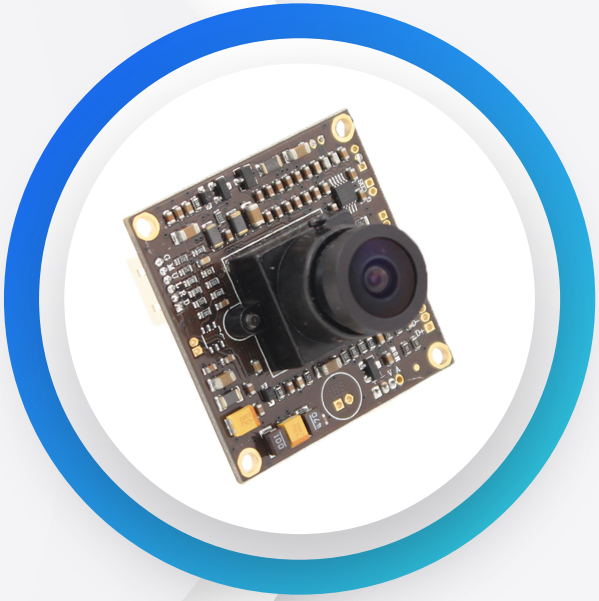
Camera Intrinsic Parameter Estimation

Your handy assistant for your robotics projects

Camera intrinsics

are the internal parameters of a camera that describe how it maps 3D points onto the 2D image plane



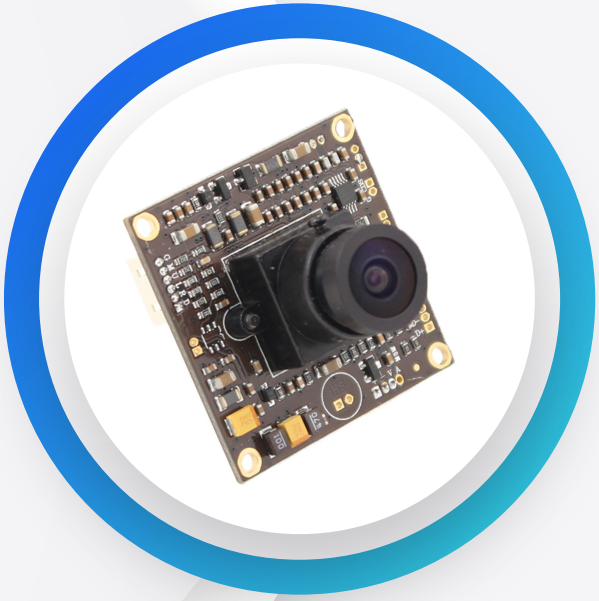
Camera intrinsics

$$K = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

f_x, f_y — focal lengths in pixels along the x & y axes

c_x, c_y — coordinates of the principal point

s — skew between the axes



Existing tools

01

CalibDB website

complex procedure for creating photos for the dataset

02

ROS

not every robotics engineer uses this ecosystem

Our proposed solution

A convenient calibration tool using a checkerboard pattern

Upload a ready-made dataset or
Create a dataset online

Calibration without a checkerboard pattern

If you don't want to use a pattern

Contribution



Ruslan Gatiatullin

Zhang algorithm implementation



Sofia Pushkareva

self calibration algorithm implementation



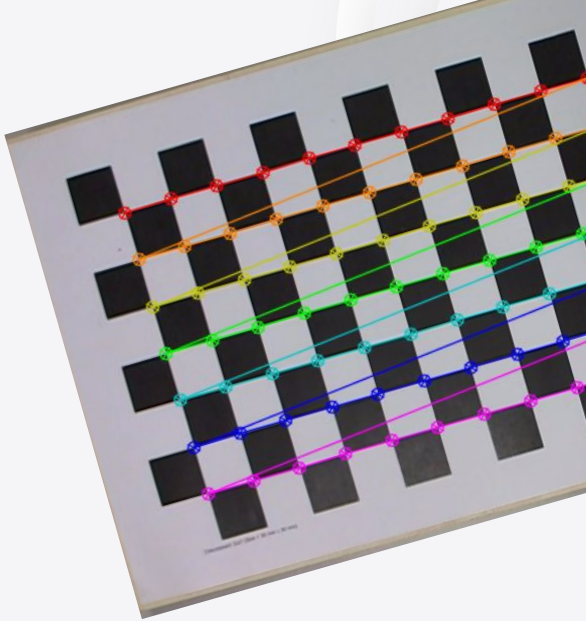
Anna Belyakova

algorithms debugging, frontend, writing reports

01

Zhang algorithm

Zhang algorithm



01

Data

Capture chessboard images and detect corner points on a plane

02

Projection Model

Model projection of planar points; projection becomes a homography

$$x_I = P\tilde{X}_w$$

$$P = KM_{Ext}$$

Zhang algorithm

03

Homography Estimation

Compute homography H from 2D correspondences via SVD.

$$H = \text{last column of } V^T \text{ (SVD)}$$

Zhang algorithm

$$\tilde{x} = x(1 + q_1 r^2 + q_2 r^4)$$
$$\tilde{y} = y(1 + q_1 r^2 + q_2 r^4)$$

04

Intrinsic Matrix

Use constraints from homographies to solve for intrinsics.

05

Distortion & Refinement

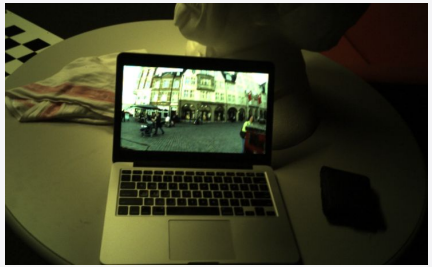
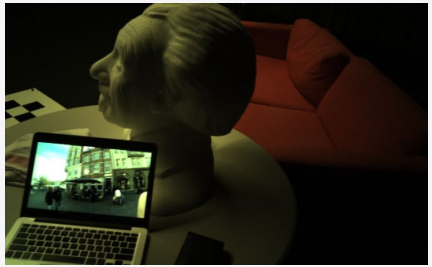
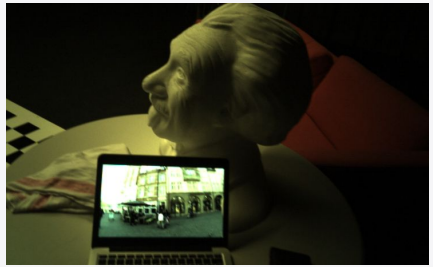
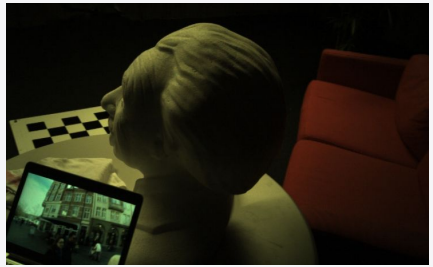
Fit distortion and refine all parameters by minimizing reprojection error.

02

Self calibration algorithm

Self calibration algorithm

1. Collect frames & Detect features
2. Match points between frames
3. Estimate the focal length
4. Recover camera motion & build sparse 3D
5. Final optimization

Results

	RMSE	Runtime	Implementation
Zhang algorithm	0.19	< 1 min	feasible
Self calibration	0.89	~ 6 min	complex


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